

Chemistry Exam-03 — Per-Exam Overview

Chapter 6: Stoichiometry | Seemab | Attempt Date: 2026-05-12

SINGLE ATTEMPT

57.5 / 65

Corrected rubric: 12 + 33 + 20 = 65 marks

88.5%

A+

Section A — MCQs

12/12

100% — Perfect

Section B — Short Qs

32/33

97.0%

Section C — Long Qs

13.5/20

67.5%

Marks Lost

7.5

out of 65

Section Breakdown

Section A — MCQs (12 marks) 12/12 = 100%

Section B — Short Questions (33 marks) 32/33 = 97.0%

Section C — Long Questions (20 marks) 13.5/20 = 67.5%

Overall (65 marks) 57.5/65 = 88.5%

Question-by-Question Breakdown

Question	Variant	Topic	Mark	Out of	%
Sec A	All 12	MCQs	12	12	100%
Q2(i)	Main	Empirical + molecular formula definitions	3	3	100%
Q2(ii)	Main	Molecular mass of glucose	3	3	100%
Q2(iii)	OR	Compare emp vs mol formula; two quantities	3	3	100%
Q2(iv)	Main	Mass of 0.25 mol CO ₂	3	3	100%
Q2(v)	Main	Molecules in 0.5 mol CH ₄	3	3	100%
Q2(vi)	Main	Chemical formulas NaCl, Mg ₃ N ₂ , K ₂ SO ₄	3	3	100%
Q2(vii)	OR	Balance Na+H ₂ O equation	3	3	100%
Q2(viii)	OR	Atoms in 0.2 mol Al	3	3	100%
Q2(ix)	OR	Ionic equation for HCl+NaOH	2	3	67%
Q2(x)	OR	Mass of 1.2 mol K; 0.25 mol steam	3	3	100%
Q2(xi)	OR	Caffeine empirical formula + mol mass	3	3	100%
Q3(i)	OR	Emp/mol defs + Aspirin/Vinegar/TNT	4	5	80%
Q3(ii)	OR	Moles of He/C/S; mass of H ₂ /H ₂ SO ₄	4	5	80%
Q3(iii)	OR	Ionic equation definition + 3 net ionic eqs	2.5	5	50%
Q3(iv)	Main	K+H ₂ O: balance, moles, atoms of K	3	5	60%

Strengths and Areas for Improvement

Strengths

- **MCQ mastery continues:** 10th consecutive perfect 12/12. Total MCQ record: 131/132 (99.2%).
- **Section B near-perfect (32/33):** 10 of 11 short questions fully correct — only ionic notation gap cost 1 mark.
- **Mass-mole interconversions automatic:** Every mass=moles×M and moles=mass÷M calculation across Sections B and C performed correctly and without hesitation.
- **Balancing equations:** Both Q2(vii) and Q3(iv)(i) balanced correctly with state symbols.
- **Chemical formula from ions:** Cross-charge method applied perfectly for all three compounds in Q2(vi).
- **Large molecular mass calculations:** Caffeine (194 amu), TNT (227.04 amu), H₂SO₄ (98.016 amu) all computed correctly — multi-element computations handled cleanly.

Priority Areas for Revision

- **Atoms = moles × N_A (not molar mass × N_A) — Q3(iv)(iii): −2 marks.** Same error in every Chemistry exam. Three-step chain: mass ÷ M → moles; moles × N_A → atoms. The molar mass M is only used once (step 1).
- **Ion charges and state symbols in ionic equations — Q2(ix) −1 mark, Q3(iii) −2 marks.** H⁺(aq) not H; Cl[−](aq) not Cl; Mg²⁺(aq) not Mg; SO₄^{2−}(aq) not SO₄. An ionic equation with no charges is just a symbol equation.
- **Net ionic — H₂ is a gaseous product, not a reactant — Q3(iii)(i) and (iii): −2 marks.** Metal + acid → Metalⁿ⁺(aq) + H₂(g). The H₂ appears after the arrow.
- **C = 12 g/mol (not 14) — Q3(ii)(a)(ii): −1 mark.** N = 14 g/mol. Flash card needed.
- **Empirical formula validity check — Q3(i)(b): −1 mark.** Dividing subscripts must produce whole numbers. Check: does 9÷2 = whole number? No (4.5) → GCD ≠ 2. Vinegar CH₃COOH = C₂H₄O₂ → divide by 2 → CH₂O.

vs. Previous Chemistry Attempts

Exam	Date	Chapters	Score	%	Sec A	Sec B	Sec C
Chem exam-01	2026-04-12	Ch 1-2	64/65	98.5%	12/12	31/33	~21/20
Chem exam-02	2026-04-17	Ch 3-5	59.5/65	91.5%	12/12	29/33	18.5/20
Chem exam-03	2026-05-12	Ch 6	57.5/65	88.5%	12/12	32/33	13.5/20

Key observation: Section B improved exam-over-exam (29 → 32/33). Section C dropped sharply (18.5 → 13.5/20) — driven entirely by ionic equation errors in Q3(iii) and the recurring atoms=moles×N_A chain error in Q3(iv). Chemistry average now: **92.8%** across three exams.